25900398

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Course (DSC) - MG3DSCFTQ200 – FOOD PRESERVATION

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap) Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions Answer any 5 questions Each question carries 2 marks

1.How does food preservation prevent microbial growth? [U] / [CO1]
Preservation usually involves preventing the growth of bacteria, fungi (such as yeasts), and other microorganisms, as well as retarding the oxidation of fats which cause rancidity. A number of methods of prevention can be used that can either totally prevent, delay, or otherwise reduce food spoilage.

2.What are the different types of refrigerant? [K] / [CO2]
The most common refrigerants
include Chlorofluorocarbons (CFCs), Hydrochlorofluorocarbons (HCFCs), Hydrofluorocarbons (HFCs), and Natural Refrigerants. Freon or halocarbons are synthetically produced.

3.What is the role of foam in foam mat drying? [U] / [CO3]
Foam plays a crucial role in the drying process of foam mats by enhancing moisture removal and improving drying efficiency.

4.Write in short on water thawing [K] / [CO3, CO4]
Water thawing refers to the process of ice or frozen water transitioning back to its liquid state. This process occurs when the temperature of the ice rises above its freezing point (0°C or 32°F).

5. Develop a note on hypertonic solution used in osmotic dehydration[A] / [CO2, CO5]
Hypertonic solutions are commonly used in osmotic dehydration processes to remove moisture from food products. This method enhances preservation and concentrates flavors**.**

6.What is the impact of moisture control on food shelf life? [K] / [CO1] Moisture control significantly affects food shelf life by preventing spoilage and extending freshness.

7.What is syruping and brining? How is it done [U] / [CO2, CO4]
Syruping is a preservation and flavoring technique that involves soaking or cooking fruit in a sugar syrup. It’s used to preserve fruit, candy fruit pieces, or create dessert toppings. Brining is the process of soaking food—usually meat, poultry, or vegetables—in a saltwater solution.
It helps retain moisture, improve texture, and enhance **[2x5 = 10]**

Part B

Short Essay Questions Answer any 5 questions Each question carries 4 marks

8.List out the advantages and disadvantages of osmotic dehydration[K] / [CO2]

Osmotic dehydration is a process where water is partially removed from fruits, vegetables, or other food materials by immersing them in a hypertonic solution (such as sugar or salt solution). It's often used as a pre-treatment before drying, freezing, or other preservation methods.

Aspect	Advantages	Disadvantages
Product Quality	Better color, flavor, and nutrients retained	Possible solute taste alteration
Processing	Gentle, low-temperature process	Time-consuming
Energy Use	Reduces drying energy	High solute consumption
Shelf Life	Lower water activity	Nutrient leaching possible
Environmental Impact	—	Waste solution disposal issues

9.Write short notes on (1)Plate freezing (2)Immersion freezing[K] / [CO3]

Plate freezing is a method used to freeze food products quickly and efficiently. It involves placing food items between two metal plates that are cooled to low temperatures. Immersion freezing is a process in which ice forms within a supercooled liquid due to the presence of solid particles that act as nucleation sites.

10.Write a brief note on radappertization, radurization and radicidation? [U] / [CO2]

Radappertization- A high-dose irradiation process (usually 30–40 kGy) used to achieve commercial sterility of food.

Radurization: A pasteurization-level irradiation process (typically 0.75–10 kGy).

Radicidation : An irradiation process (around 2–8 kGy) aimed at destroying specific non-spore-forming pathogenic microorganisms.

11.Enumerate the preservation method used in the killing of microorganisms? [K] / [CO1]

Pasteurization ,sterilization, Irradiation, cold treatment

12.Explain in detail on flash 18 process? [U] / [CO2] The Flash 18 process is a continuous canning/sterilisation method in which the product is brought up to sterilising temperature prior to filling, and the filling and sealing are done under pressure (≈ 18 psi above atmospheric) so that the product remains above its normal boiling point. This enables sterilisation at higher temperature and shorter time, improving product quality

13. Explain the different types of hurdles used in combined preservation of foods [K] / [CO2]

1. Temperature Control- Refrigeration, Freezing, Heat Treatment: Includes pasteurization and sterilization to kill pathogens. 2. Water Activity Reduction-Dehydration, Salting, Sugaring 3. pH Control 4. Chemical Preservatives 5. Modified Atmosphere Packaging (MAP) 6. Biological Control

14.What do you mean by exhausting of cans? List the methods and adavantages of exhausting **[K] / [CO2,CO4]**

Exhausting is a unit operation in which practically all the air is removed from the contents of the can before sealing.

Methods of exhausting- Heat, steam, cold and mechanical exhausting

Adavantages of exhausting –Prevents oxidation of food components, Reduce corrosion of metal cans, Ensure formation of proper vacuum

[4x5 = 20]

Part C

Long Essay Questions Answer any 2 questions Each question carries 10 marks

15.Write an essay on the following types of pasteurization!) LTLT 2) HTST 3) UHT [U] / [CO2]

16. Define irradiation. Give an account of the major applications of irradiation in food Industry [A] / [CO2]

- Irradiation is the process by which an object is exposed to radiation. An irradiator is a device used to expose an object to radiation, most often gamma radiation, for a variety of purposes. Food irradiation has various practical applications, including sprout inhibition in tubers and bulbs, delay in fruit ripening, and decontamination of spices and other food products.

17. Write an essay on the types of sterilizers used in food industries[U] / [CO2]

Hydrolock, hydrostatic pressure sterilizer, Flame sterilizer

18.State the principle of hurdle technology. Discuss the application of hurdle technology in 1) Moisture foods 2) Chilled foods 3) fermented foods [K] / [CO2]

Hurdle technology is defined as a technique that combines existing and novel methods for food preservation to improve quality and inhibit microbial growth. State the application in moisture foods, chilled foods and fermented foods

[10x2 = 20]